Know-how sharing with stochastic innovations

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Abstract. We provide a model of know-how sharing between competing firms in which each of two firms gets a stochastic innovation in its stock of know-how in every period. Separately considering the cases when innovations are indivisible and divisible, we examine the nature of the subgame perfect sharing agreements that can obtain. We discover that both stochasticity and indivisibility undermine the ability to support sharing. Furthermore, we find that there are equilibria in which know-how sharing can be intermittent and that small innovations are more likely to be shared than large ones, when innovations are divisible but not necessarily when they are indivisible. JEL Classification: O30, O33

Partage du savoir faire quand les innovations sont stochastiques. Les auteurs proposent un modèle de partage du savoir-faire entre entreprises concurrentes dans lequel chacune des deux entreprises obtient une innovation stochastique dans son stock of savoir-faire à chaque période. En considérant séparément les cas où les innovations sont divisibles et non-divisibles, on examine la nature des accords de partage parfait qui peuvent se produire dans le sous-jeu. On montre que la stochasticité et l'indivisibilité minent la possibilité de maintenir le partage. De plus, on découvre que des solutions d'équilibre avec partage de savoir-faire peuvent jouer par intermittence, et qu'on est davantage susceptible de partager les fruits des petites innovations plus que des grandes quand les innovations sont divisibles, mais pas nécessairement quand elles sont indivisibles.

1. Introduction

The phenomenon of know-how sharing among firms that are competitors has been documented by a number of analysts. Two of the most interesting cases come from the steel industry. In his study of the evolution of the blast furnace between 1859

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and 1875, Allen (1983) found pervasive know-how sharing, which he dubbed 'collective invention.' More recently, in his study of the steel minimill industry in the United States, von Hippel (1988) found that rival firms routinely exchanged technical information. This sometimes went so far as firms setting up entire plants for their rivals. Carter (1989) explores some of the features that make trade in know-how different from ordinary trade.

To explain what he observed in the steel minimill and other industries, von Hippel (1988) proposed a theory of know-how sharing based on the idea that it is cost reducing. In any one period, two firms can be seen as playing a game in which they choose to share or not to share the innovations to the know-how that arise in that period. In these circumstances, it is reasonable to suppose that strategy profile (*not share*, *not share*) is a dominant strategy equilibrium in each period, while strategy profile (*share*, *share*) Pareto-dominates it. Further, if we assume that the game does not change from period to period, know-how sharing can be seen as taking place in a supergame, and standard results can be invoked to explain such sharing. In particular, if firms do not discount future profits too heavily, it is possible to support (*always share*, *always share*) as a subgame perfect Nash equilibrium by using trigger strategies that call for reversion to the dominant strategy equilibrium in the event of defection. Baumol (1993) both surveys the existing empirical evidence on the phenomenon of know-how sharing and articulates and extends the theory proposed by von Hippel (1988). In this, as in a related paper (Eaton and Eswaran 1997), we build on the pioneering efforts of von Hippel and Baumol.[1]

At the heart of this theory of know-know sharing is an exogenous process that drives innovations in know-how. This innovation-generating process is intended to loosely capture learning-by-doing. An important feature of the real process of learning-by-doing that is not incorporated in the theory sketched by von Hippel (1988) and Baumol (1993) is stochasticity. From the perspective of a firm's managers, innovations to know-how are surely stochastic. One objective in this paper is the exploration of how the properties of a stochastic innovation-generating process affect the possibilities for equilibrium know-how sharing.

An important property of know-how is indivisibility. If A discovers some clever idea, she can share it with B or keep it from B, but A may not be able to share a part of her idea. Another of our objectives is to explore the implications of indivisibility for equilibrium know-how sharing. We do this by comparing the set of equilibrium know-how sharing agreements when know-how is indivisible with the corresponding set when know-how is assumed to be perfectly divisible.

By its very nature, know-how is non-rivalrous, and we assume that it is excludable. Thus, firms can choose to share their know-how or to not share it. In addition, we assume that patenting and/or licensing is not feasible. Thus, standard property

1 In a different context, Gallini (1984) has demonstrated that it may be in the interest of a firm to share its technology with a rival, if by doing so the rival is deterred from doing R&D that would prove damaging to the former. The underlying motivation for know-how sharing is quite different in this scenario.

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rights arrangements cannot be used to control and direct the diffusion of know-how. We make these assumptions because we think they are appropriate for many real-world situations and because they allow us to isolate the phenomenon of interest: voluntary know-how sharing. The central question of interest here is this: Under what circumstances can voluntary know-how sharing effect an efficient diffusion of know-how?

In our model, in each period two risk neutral firms get independent draws of know-how from a common probability distribution. A firm's profit in any period is an increasing function of its own stock of know-how and a decreasing function of the stock of know-how of its competitor. Hence, in any period, (*not share*, *not share*) is a dominant strategy equilibrium. But the marginal effect on profit of own and competitor's stocks of know-how are such that (*always share*, *always share*) is Pareto-efficient. We look at sharing agreements that are supported by reversion to the no-sharing equilibrium.

Our analysis generates two results that are especially interesting. First, we find that stochasticity inhibits the ability to cooperate, especially in situations where the distribution of innovations has a long, thin, right-hand tail; in these situations, full sharing is difficult to support because a firm that gets an exceptionally valuable, and rare, draw from the distribution must anticipate that the other firm will not be in position to reciprocate for a very long time. More generally, a mean-preserving spread in the distribution of innovations renders full know-how sharing more difficult to support. Second, an interesting feature of the equilibrium of our model is that, when know-how is indivisible, it is possible that sharing is intermittent even when a sharing agreement is in effect. That is, we will observe periods during which no information is being exchanged and cooperation has not broken down: the sharing agreements are still in force. This stands in contrast with the usual scenarios in oligopoly, where absence of cooperation for any length of time in a supergame indicates that a punishment phase is in effect.

We find that relative to the case of perfect divisibility, indivisibility of know-how inhibits the ability to cooperate. We uncover a surprising result here: when innovations are indivisible, it is very difficult to support a small amount of cooperation and impossible in the limit as the amount of cooperation goes to zero. Together, these results mean that, when innovations are stochastic and know-how is indivisible, it may be impossible to support any cooperation where full cooperation in the corresponding deterministic scenario may well have been possible.

In the context of oligopoly theory, it is well known that stochasticity makes collusion difficult, as seen in Green and Porter (1984) and Rotemberg and Saloner (1986). The mechanism in Green and Porter (1984) is very different from that proposed here. There, randomness in the demand facing the industry makes it difficult for firms to ascertain whether cheating has taken place; so, as a way of enforcing collusion, firms automatically go into a punishment mode (a price war) for some length of time if the price falls below some critical level. Closer in spirit to our analysis is that of Rotemberg and Saloner (1986), who argue that the incentive for a firm to cheat is greater during booms than during recessions. However, there

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are peculiarities associated with know-how (indivisibility, in particular) that interact with stochasticity in a manner that precludes this argument from being carried over to the context of know-how sharing. As we demonstrate, it is with perfectly divisible know-how that stochasticity yields results similar to those in oligopoly theory. It might be thought that results from the literature on Folk Theorems are applicable to our model (see, for example, Fudenberg, Levine, and Maskin 1994). This is not the case, since our model, in fact, is not a repeated game. In each period, the action sets of the firms, and hence their payoff functions, are determined by stochastic realizations of know-how, so that what might loosely be called the stage game in our model changes from period to period.

There are a number of other situations to which our model is applicable. Informal mutual insurance schemes of the sort that are widespread in countries with undeveloped insurance markets have some (but not all) important features in common with our model. The informal pooling of spare parts by firms that are competitors in their output markets is another phenomenon that our model helps to explain. Thus, while exploring the specific issue of know-how sharing, we suggest insights that are potentially relevant in more general scenarios.

The paper is organized as follows. In the next section, we spell out the basic model we use throughout. In section 3, we examine the incentive compatibility conditions that must be satisfied by know-how sharing agreements, and in section 4, we determine the optimal form of sharing agreements when know-how is indivisible. In sections 5 and 6, we derive the results our model generates for the polar cases where know-how is indivisible and perfectly divisible, respectively. We present our concluding remarks in the last section.

2. The model

Assume that there are two identical firms repeatedly confronting each other over an infinite number of periods. In every period each firm receives an exogenous, private innovation which impinges on its own profit and that of its rival. We shall take it that the larger the innovation of a firm, the better it is for the firm and, in the absence of sharing, the worse it is for its rival. The innovations for both firms are drawn from a time-invariant probability distribution, and they are assumed to be independent. Each firm's innovation, x, has a density function $f(x)$ defined on the support $[\underline{x}, \bar{x}]$, denoted hereafter by the set Y. We assume that $0 \leq \underline{x} < \bar{x} < \infty$, and that $f(x) > 0$ on the open interval $(\underline{x}, \bar{x})$. We denote the mean value of the innovations by μ:

$$\mu = \int_{\underline{x}}^{\bar{x}} xf(x)\,dx.$$

We denote firm i's innovation in any period by x_i, $i = 1,2$. (In what follows, firm j will denote the firm other than i.) The density function, $h(x_i, x_j)$, associated with the joint distribution is given by $h(x_i, x_j) = f(x_i)f(x_j)$.

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Let g_i denote the transfer ('gift') of innovation received *by* firm i from firm j, and g_j that received *by* firm j from firm i. We assume that sharing entails zero transactions cost if undertaken in the period in which the innovation materializes and that current innovations are infinitely costly to share in a later period. Equivalently, we may assume that last period's information has no value this period. This allows us to ignore the possibility that firms share in some period an innovation that they acquired but witheld in an earlier period. We can then assume that $0 \leq g_i \leq x_j$ and $0 \leq g_j \leq x_i$. For simplicity, we further assume that the payoff to each firm depends only on the innovations and gifts (received and given) of this period: there is no stock dependence.[2] The payoff to firm i in any period is denoted by the function $\Pi^i(x_i, x_j; g_i, g_j)$, continuous and differentiable in all its arguments. We assume that the partial derivatives of this function, denoted by the respective subscripts, have the following signs:

$$\begin{aligned} &\Pi^i_1(x_i, x_j; g_i, g_j) > 0; \qquad \Pi^i_2(x_i, x_j; g_i, g_j) < 0 \\ &\Pi^i_3(x_i, x_j; g_i, g_j) > 0; \qquad \Pi^i_4(x_i, x_j; g_i, g_j) < 0. \end{aligned} \tag{1}$$

These inequalities say that, from the point of view of firm i: larger innovations of x_i are better; larger innovations of x_j are worse; larger gifts received from firm j are better; and larger gifts made to firm j are worse. However, we shall posit that small equal exchanges between firms are necessarily beneficial on balance; that is,

$$\Pi^i_3(x_i, x_j; 0, 0) + \Pi^i_4(x_i, x_j; 0, 0) > 0 \qquad \text{for any } (x_i, x_j). \tag{2}$$

The last inequality in (1) is the source of the potential for cheating: in a one-period scenario, each firm will find it rational to withold information from its rival. However, the inequality in (2) indicates that a small reciprocal exchange of information, on balance, benefits both firms. This provides the long-term motivation for sharing.

Define the function

$$\Psi^i(x_i, x_j, g_i) \equiv \Pi^i(x_i, x_j; g_i, 0) - \Pi^i(x_i, x_j; g_i, x_i), \tag{3}$$

which represents the immediate *gain* to firm i when it transfers no information to its rival relative to when it transfers all of its information. From the last inequality in (1), it follows that $\Psi^i(x_i, x_j, g_i) > 0$ when $x_i > 0$. We assume that

$$\frac{\partial \Psi^i(x_i, x_j, g_i)}{\partial x_i} \geq 0; \quad \frac{\partial \Psi^i(x_i, x_j, g_i)}{\partial g_i} \leq 0. \tag{4}$$

2 We are following Eaton and Eswaran (1997) in invoking these simplifying assumptions. However, it is possible to accommodate the case where this period's realizations and gifts contribute to stocks, which, in turn, determine the payoffs. The associated analysis for the special case where the payoffs are linear in the stocks may be obtained from the authors. The essential points of this paper are not changed by the introduction of this complication, and this is why we opt for the simpler model without stocks.

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The first inequality in (4) says that the immediate gain to firm i from witholding information is non-decreasing in the size of its innovation. That is, witholding information is more profitable (in the current period) when there is a lot to withold. The second inequality says that the gain to firm i from witholding information is non-increasing in the size of the gift it receives from its rival. Loosely speaking, there are 'non-increasing returns' to cheating in the amount of know-how the firm has access to. That is, the larger the gift received from the rival, the less the advantage a firm's own innovation bestows.

We suppose, first, that innovations are indivisible. Accordingly, in any period a firm can either share its innovation or not share it, but partial sharing is not possible. For purposes of comparison, in a subsequent section we replace this assumption with its opposite extreme: innovations are perfectly divisible.

A *sharing agreement* is a closed set $\Omega \subseteq \Upsilon$ of innovations: the agreement calls for firm i to share its innovation x_i with firm j if and only if $x_i \in \Omega$. We restrict ourselves to symmetric sharing agreements for the two firms. The set Ω (which we will loosely refer to as the agreement set) is not necessarily convex; that is, it may have 'holes' in which innovations are not expected to be shared. Let Λ denote the complement of Ω; Λ is the set of innovations that are not called for by the agreement to be shared.

Support for the sharing agreement is provided by a grim trigger strategy of the sort first used by Friedman (1971) that calls for reversion to no sharing in the event of defection. The strategy profile (*never share, never share*) clearly constitutes a subgame perfect equilibrium (SPNE) since, by (1), each firm's one-period payoff is decreasing in the gift it gives the rival. Notice that the reversion strategy posited here entails the severest possible punishment for defection in this game.

The timing of the events is as follows. Prior to the first period of an infinitely repeated game, the firms come to an understanding regarding a sharing agreement. In each period of the game, innovations materialize and are privately revealed to the two firms. Then, each firm has the opportunity to share its innovation with the other firm. Finally, payoffs for the current period are realized.

Since firm i knows the function $\Pi^i(x_i, x_j; g_i, g_j)$, it can invert this function to infer the innovation of its rival in this period. Hence, at the beginning of a period, each firm knows whether the other has violated the sharing agreement in the previous period. Since the innovations are privately received by each firm and can be only indirectly inferred by the rival, we assume that the firms cannot write explicit contracts that are binding in a court of law.

We can think of a sharing agreement in terms of the reciprocal expected gifts that are exchanged under it. Let G denote the expected gift one firm receives from (and gives to) the other under sharing agreement Ω:

$$G_\Omega = \int_\Omega x f(x)\, dx. \tag{5}$$

From an economic perspective, this *reciprocal gift* – as G_Ω will be referred to hereafter – is of considerable interest. Because the reversion strategy that supports

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sharing agreements in our model entails the severest possible punishment for defection, our analysis fully delineates the set of reciprocal gifts that can be supported as an SPNE of our sharing game. Since full sharing will be Pareto optimal in our model, it would be interesting to delineate the circumstances in which full sharing (that is, when G is equal to μ) is supportable.

For brevity, we introduce the following notation for arbitrary sets $A \subseteq \Upsilon, B \subseteq \Upsilon$:

$$\langle \Pi^i(x_i, x_j; g_i, g_j) \rangle_{A,B} \equiv \int_A \int_B \Pi^i(x_i, x_j; g_i, g_j) h(x_i, x_j)\, dx_i\, dx_j, \tag{6}$$

with the convention that the left-hand integral is that for x_i over the set A and the right hand integral is that for x_j over the set B. The expected one-period, ex ante payoff to firm i under the agreement, $E\Pi^i_\Omega$, may then be written as

$$E\Pi^i_\Omega = \langle \Pi^i(x_i, x_j; x_j, x_i) \rangle_{\Omega,\Omega} + \langle \Pi^i(x_i, x_j; 0, x_i) \rangle_{\Omega,\Lambda} + \langle \Pi^i(x_i, x_j; x_j, 0) \rangle_{\Lambda,\Omega} + \langle \Pi^i(x_i, x_j; 0, 0) \rangle_{\Lambda,\Lambda}. \tag{7}$$

The four terms on the right-hand side of (7) represent, respectively, the contributions to the expected payoff of firm i in the current period when the innovations are such that the agreement requires (i) both firms to share; (ii) firm i to share but not firm j; (iii) firm j but not firm i to share; and (iv) neither firm to share.

Consider any two sharing agreements characterized, respectively, by the sets Ω and $\tilde{\Omega}$, such that $\Omega \subseteq \tilde{\Omega} \subseteq \Upsilon$. We shall assume that

$$E\Pi^i_{\tilde{\Omega}} \geq E\Pi^i_\Omega, \tag{8}$$

with the inequality being strict if the Lebesgue measure of $\tilde{\Omega}$ is strictly greater than that of Ω. This is a reasonable requirement: it posits that the ex ante expected payoff to a firm does not decrease if the agreement set is expanded. In particular, full sharing is Pareto optimal. The assumptions stated in (1), (2), (4), and (8) will be maintained throughout this paper. (It may be noted as an example that a simple, linear payoff function to be considered later in this paper readily satisfies all of these assumptions.) It may be noted that, when $\underline{x} = 0$, (2) may be derived from (8) by choosing Ω to be the null set and $\tilde{\Omega}$ to be a set of non-zero measure comprising small innovations.

3. Incentive compatibility

In view of (8), any agreement that generates a positive reciprocal gift, G_Ω, will be strictly preferred to no sharing. Hence, both firms would willingly participate in an enforceable agreement Ω with $G_\Omega > 0$. We restrict attention to such agreement sets (that is, those with non-zero measure).

Since the sharing of a firm's innovation with the rival reduces the former's payoff in the current period, however, each firm has an incentive to cheat when its innovation falls in the set Ω. As mentioned earlier, each firm's innovation can be inferred

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by its rival at the beginning of the next period. Thus, any gains from cheating accrue for only one period. In the event of cheating, both firms will revert to a punishment mode that entails no sharing for ever.

Let δ ($0 \leq \delta < 1$) denote the common discount factor of the two firms. Innovations $x_i \in \Lambda$ do not raise issues of cheating, since such innovations are not called upon by the agreement to be shared. Suppose, however, that firm i has obtained an innovation $x_i \in \Omega$ in this period and does not yet know firm j's innovation. In the event that neither firm cheats, firm i's expected present value payoff, given its current period innovation, is

$$\int_{\Omega} \Pi^i(x_i, x_j; x_j, x_i) f(x_j)\, dx_j + \int_{\Lambda} \Pi^i(x_i, x_j; 0, x_i) f(x_j)\, dx_j + \frac{\delta}{1-\delta} E\Pi^i_{\Omega}. \tag{9}$$

If, however, firm i decides to cheat in the current period, its expected present value payoff, given its current period innovation, is

$$\int_{\Omega} \Pi^i(x_i, x_j; x_j, 0) f(x_j)\, dx_j + \int_{\Lambda} \Pi^i(x_i, x_j; 0, 0) f(x_j)\, dx_j + \frac{\delta}{1-\delta} E\Pi^i_{\Phi}, \tag{10}$$

where the subscript Φ in the third term denotes the null set, and thus

$$E\Pi^i_{\Phi} \equiv \langle \Pi^i(x_i, x_j; 0, 0) \rangle_{\Upsilon, \Upsilon} \tag{11}$$

denotes the one-period ex ante expected payoff to firm i when there is no sharing agreement in place.

If firm i is not to cheat, the expression in (9) must be at least as large as that in (10). This incentive compatibility constraint may be written

$$P^i_{\Omega}(\delta) \geq T^i_{\Omega}(x_i), \tag{12}$$

where $P^i_{\Omega}(\delta)$, the expected 'punishment' to firm i for cheating, is given by

$$P^i_{\Omega}(\delta) = \frac{\delta}{1-\delta} [E\Pi^i_{\Omega} - E\Pi^i_{\Phi}], \tag{13}$$

and $T^i_{\Omega}(x_i)$, firm i's 'temptation' to cheat, is given by

$$T^i_{\Omega}(x_i) = \int_{\Omega} \Psi^i(x_i, x_j; x_j) f(x_j) dx_j + \int_{\Lambda} \Psi^i(x_i, x_j; 0) f(x_j) dx_j, \tag{14}$$

where the function Ψ^i is defined in (3). The first (second) term on the right-hand side of (14) represents the expected gain to firm i of cheating when firm j's innovation is such that it is obliged (not obliged) by the agreement to share its innovation with firm i. Both the integrands on the right-hand side of the expression in (14) are non-decreasing in the magnitude of the innovation x_i by the first inequality in (4). The punishment for cheating is simply the present value of the difference between the expected future profit to firm i with and without sharing. This is obviously

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independent of the current-period innovations of the firms. Let X denote this largest element of Ω. Since the temptation to cheat is greatest when firm i's innovation is equal to the largest element of Ω, from (12) the incentive compatibility constraint that is necessary and sufficient for Ω to be supportable is given by

$$P^i_\Omega(\delta) \geq T^i_\Omega(X). \tag{15}$$

This incentive compatibility condition is the same in all periods, since the game under consideration is stationary.

4. The optimal form of sharing agreements

We now will invoke the Pareto criterion to characterize the optimal form of sharing agreements. As mentioned earlier, the set Ω does not necessarily include all innovations smaller than X; that is, the set Ω is not necessarily convex. We now demonstrate that Pareto-optimal agreements must be characterized by convex sharing sets. We call sharing agreement Ω *form-optimal* if it has the following form: $\Omega = \{x : \underline{x} \leq x \leq X\}$, where $\underline{x} < X \leq \bar{x}$. That is, the set of shared innovations is form optimal if it can be described by a real number X lying in the support of the distribution such that an innovation is to be shared if and only if it is no larger than X. The essential feature of a form-optimal agreement is that if, according to the agreement, a particular innovation is to be shared then all smaller innovations are also to be shared: the set of shared innovations is convex. In proposition 1, below, we show that Pareto optimality requires form optimality.

PROPOSITION 1. *If a supportable sharing agreement is not form optimal, there is a supportable form-optimal agreement that (weakly) Pareto-dominates it.*

To prove this proposition, suppose Ω is a non-form-optimal sharing agreement, with X as its largest element. Consider what happens when the sharing set is expanded to $\tilde{\Omega} \supseteq \Omega$ while maintaining X as the largest element of $\tilde{\Omega}$ also. We can write $\tilde{\Omega} = \Omega \cup \Theta$, where Ω and Θ are mutually exclusive sets. The change in the expected current period payoff to firm i for cheating, $\Delta T^i(x_i)$, is

$$\Delta T^i(x_i) = \int_\Theta [\Psi^i(x_i, x_j; x_j) - \Psi^i(x_i, x_j; 0)] f(x_j) dx_j,$$

which is the expected gain in cheating to firm i when firm j is obliged to share its innovation in Θ less the expected gain when firm j is not thus obliged. Since, by the second inequality in (4), the gain to a firm from cheating is non-increasing in the gift received from its rival (owing to 'non-increasing returns' to information), the term in the square bracket in the integrand is always non-positive. Hence, $\Delta T^i(x_i) \leq 0$; that is, the temptation to cheat does not increase when the sharing agreement is expanded from Ω to $\tilde{\Omega}$.

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The change in the punishment to cheating, $\Delta P^i(\delta)$, when Θ is added to the sharing set is given by

$$\Delta P^i(\delta) = P^i_{\tilde{\Omega}}(\delta) - P^i_{\Omega}(\delta),$$

which, from (8) and (13), is non-negative, since the expected payoff under the agreement set Ω is no larger than that under the agreement set $\tilde{\Omega}$. Thus, if the agreement set Ω is supportable, so is $\tilde{\Omega}$. Furthermore, since $E\Pi^i_{\tilde{\Omega}} \geq E\Pi^i_{\Omega}$, $\tilde{\Omega}$ (weakly) Pareto-dominates Ω, and the dominance is strict if the Lebesgue measure of $\tilde{\Omega}$ is strictly greater than that of Ω.

In general, we can conceive of the sharing agreements being asymmetric, despite the firms' being identical. In the interest of simplicity, however, we will restrict ourselves to symmetric agreements.[3] This is consistent with von Hippel's observation in the steel minimill industry that 'in an informal way, participants [of a know-how sharing arrangement] appeared to keep a balance in the value given and received' (1988, 82).

According to proposition 1, in effect we can learn everything there is to know about the set of supportable reciprocal gift exchanges by restricting attention to form-optimal sharing agreements. So we confine ourselves to such agreements. Given that a sharing agreement is form optimal, there is a one-to-one relationship between sharing agreements and their associated reciprocal gift exchanges, and this relationship can be expressed in two ways. First, if X is the largest element in a form-optimal sharing set, the expected gift exchange can be denoted (with a slight abuse of notation) by $G(X)$. Second, if the reciprocal gift is G, the largest element in the form-optimal agreement that yields this expected reciprocal gift can be written as $X(G)$. Obviously, $X(G)$ is the inverse of $G(X)$. The relationship between X and G is defined by the following:

$$G = \int_{\underline{x}}^{X} xf(x)\,dx. \tag{16}$$

For any G, the form-optimal sharing agreement, $\Omega(G)$, that generates this gift exchange is clearly defined by

$$\Omega(G) \equiv \{x : \underline{x} \leq x \leq X(G)\}. \tag{17}$$

We denote the complement of this set by $\Lambda(G)$.

3 When the payoffs are linear in know-how, however, it can be shown that if the sharing agreements for identical firms are not symmetric (contrary to our assumption), it can be weakly Pareto-dominated by a symmetric sharing agreement, with the dominance being strict if the agreement sets have different Lesbegue measures. The reason is that unequal reciprocal gifts dilute the incentives for sharing of the firm receiving the smaller reciprocal gift – a situation that can be remedied by making the sharing agreement symmetric.

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5. Indivisible innovations

It might appear that agreements requiring the sharing of large innovations will not likely be supported, since these require firms to give away substantial innovations that arrive infrequently. On the other hand, it might appear that agreements restricted to the sharing of small innovations are less prone to this problem and hence would be supportable. It is surprising that when innovations are indivisible, the latter conjecture is incorrect, as proposition 2 demonstrates. In fact, we shall see in this section that there are scenarios in which the ease with which form-optimal agreements can be supported increases monotonically with the size of the largest innovation in the sharing set, with full sharing agreements being the easiest to support.

PROPOSITION 2. *For any* $\delta \in [0,1)$ *and* $\underline{x} > 0$*, it is impossible to support as SPNE agreements with sufficiently small reciprocal gift exchanges.*[4]

To prove this notice that, if $G \to 0$, it follows from (16) that $X(G) \to \underline{x}$ and $\Omega \to \Phi$. Thus, from (13),

$$\lim_{G \to 0} \quad P^i_{\Omega(G)}(\delta) = 0. \tag{18}$$

The punishment for cheating on agreements with low gift exchanges is vanishingly small. This is because most of the time in the future the rival will obtain innovations exceeding $X(G)$, which it is not obliged to share in any case. Now from (14),

$$\lim_{G \to 0} \quad T^i_{\Omega(G)}(X(G)) = \int_{\underline{x}}^{\bar{x}} \Psi^i(\underline{x}, x_j; 0) f(x_j)\, dx_j$$

$$> 0, \tag{19}$$

since $\Psi^i(\underline{x}, x_j; 0) > 0$. The temptation to cheat remains strictly bounded away from zero even as the gift exchange approaches zero. Thus, the incentive compatibility condition (15) is necessarily violated in the limit $G \to 0$. Since $P^i_{\Omega(G)}(\delta)$ and $T^i_{\Omega(G)}(X(G))$ are continuous in G, it follows that there is a range of small values of G for which (15) is violated.

This result does not say that small innovations will never be shared. In fact, if sharing agreement $\Omega(G)$ is supportable, all innovations smaller than $X(G)$ will be shared. What the result does say is that if the expected reciprocal gift G associated with a sharing agreement is sufficiently small, the agreement is not supportable. That is, agreements that share only small innovations cannot be supported. Intuitively, this result is just a variation on an old aphorism: 'A bird in the hand is worth two in the bush,' because the probability that one will actually catch the two birds is small. In the present

4 In the case of linear payoffs considered below, this result holds even when $\underline{x} = 0$.

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scenario, the bird in the hand is, of course, $X(G)$, the innovation that a firm must share with its competitor if the sharing agreement is to be supportable. The birds in the bush are the future gifts that, given the agreement, the rival will share with the firm. Should the firm share $X(G)$ today to get these future gifts? None of these future gifts is larger than $X(G)$, since the sharing set is $[\underline{x}, X(G)]$, and, when G is small, the probability that the firm will actually receive a gift in any future period is quite small, since the competitor is required to share only those innovations that fall in the sharing set. According to the proposition, when G is small, not sharing $X(G)$ with the competitor has a higher value than the expected future gifts from the competitor – the bird in the hand is worth more than the birds in the bush.

5.1. An example with linear payoffs

A special case that is instructive to consider is one in which the payoff function has the following linear structure:

$$\Pi^i(x_i, x_j; g_i, g_j) = \alpha(x_i + g_i) - \beta(x_j + g_j), \qquad \alpha > \beta > 0. \tag{20}$$

When the parameter β is zero or negative, the question of sharing has no interesting economic implications, since in each period (*share, share*) is both Pareto optimal and a dominant strategy equilibrium. So we assume that $\beta > 0$. Conversely, when $\beta > \alpha$, in each period (*not share, not share*) is both Pareto optimal and a dominant strategy equilibrium, and there is no economic incentive to share innovations. So we posit that $\alpha > \beta > 0$. This linear payoff function satisfies all of the restrictions we placed earlier on the payoff functions.

The incentive compatibility condition (15) can be readily shown in this case to reduce to

$$\frac{\delta}{1-\delta}(\alpha - \beta)G \geq \beta X(G). \tag{21}$$

This inequality is easily interpreted. According to the payoff function, an increase in a firm's own innovation by one unit increases its payoff by α, while an increase in the rival's innovation by one unit decreases the firm's payoff by β. Thus, if a firm's innovation is the largest one, $X(G)$, within the agreement set, the current period gain from witholding this from the competitor is given by the right-hand side of (21). This is the largest temptation to cheat the firm has to counter. If the firm succumbs, it will lose profits in all future periods. The expected value of the gift expected to be *received and given* in any period is G, and the net profit expected in any period due to this gift exchange is $(\alpha - \beta)G$. Since this is forgone in every future period, the present value of these forgone expected profits – the punishment – is given by the left-hand side of (21). The above incentive compatibility condition may be rewritten as

$$\frac{X(G)}{G} \leq \frac{\delta}{1-\delta}\frac{(\alpha - \beta)}{\beta}. \tag{22}$$

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Inequality (22) is a necessary and sufficient condition for agreement $\Omega(G)$ to be supportable as a SPNE of the know-how sharing game. Since the benefit of cheating is proportional to $X(G)$, while the cost to cheating is proportional to G, the left-hand side of (22) is a measure of the largest benefit-cost ratio of cheating when payoffs are linear in know-how.

As we observed earlier, efficiency demands full sharing. For the form-optimal agreement that generates full sharing (that is, $\Omega = \Upsilon$), we have $G = \mu$ or $X(\mu) = \bar{x}$. Thus, (22) implies that full sharing is supportable in the linear case if and only if

$$\frac{\bar{x}}{\mu} \leq \frac{\delta}{1-\delta}\,\frac{(\alpha-\beta)}{\beta}.$$

It is apparent from the above inequality that if the upper support of the distribution, $\bar{x}$, is sufficiently large relative to the mean of the distribution, μ, it will be impossible to support full sharing. Full sharing cannot be supported for distributions with long tails.

If the innovations were certain, that is, if the probability density function $f(x)$ had all its mass at one point, then $X(G)$ would be equal to G, and the incentive compatibility condition would reduce to $1 \leq \delta(\alpha-\beta)/[\beta(1-\delta)]$. But innovations are stochastic in our model, with the result that $X(G) > G$, because G is the density-weighted average of x between $\underline{x}$ and $X(G)$. It is then clear that stochasticity undermines the ability to support voluntary sharing arrangements.[5] It is also clear that the following is a necessary condition that must hold for any sharing to be possible when innovations are stochastic:

$$1 \leq \frac{\delta}{1-\delta}\,\frac{(\alpha-\beta)}{\beta}. \tag{23}$$

We assume this condition holds.

Suppose, further, that the innovations are uniformly distributed and that the density function is given by

$$f(x) = \begin{cases} 1/(2s) & \text{for } \mu - s \leq x \leq \mu + s, \\ 0 & \text{otherwise,} \end{cases} \tag{24}$$

where it is assumed that $\mu \geq s \geq 0$. This distribution has two parameters: the mean, μ, and what we can think of as the spread parameter, s. (The parameter restrictions $\mu \geq s \geq 0$ are necessry to ensure that $x \geq 0$.) For this distribution, it can easily be shown that

$$X(G)/G = [(\mu - s)^2 + 4sG]^{1/2}/G.$$

5 This is somewhat reminiscent of the insight, in a different context, of Rotemberg and Saloner (1986) that price-fixing cartels are harder to sustain in periods of high demand than in periods of low demand.

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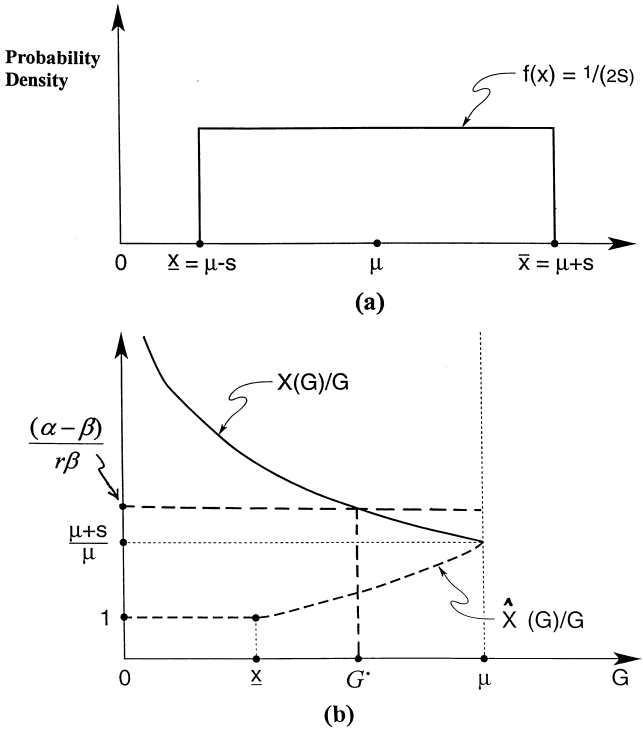


FIGURE 1 The uniform distribution

The density function for the uniform distribution is illustrated in figure 1 (a), and the function $X(G)/G$ is illustrated by the solid line in figure 1 (b). The latter function is downward sloping, is equal to $(1 + s/\mu)$ when G is equal to μ, and shoots off to infinity as G approaches 0. In the case illustrated in the figure, only reciprocal gifts larger than G^* are supportable. Two remarkable conclusions follow regarding the sharing game when the payoff function is linear and the distribution of innovations is uniform. First, if an agreement with some reciprocal gift G can be supported as an SPNE, then so can all agreements with larger expected gifts. Indeed, for any sharing agreement to be supportable, full sharing must be supportable. Second, reciprocal gifts smaller than some critical value cannot be supported. This second feature is a special case of the general result demonstrated earlier in this section.

Given what we know from more familiar oligopoly applications, these results are surprising. When firms produce substitutes and compete in quantities, it is easier to support outcomes that are closer to the Nash equilibrium of the stage game than it is to support solutions that are near the joint profit-maximizing outcome of the stage game. On the other hand, if producers of a homogeneous good compete in

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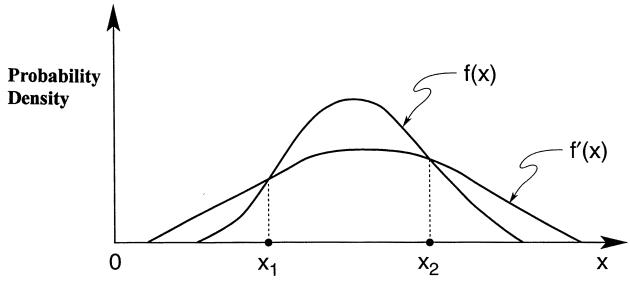


FIGURE 2 A mean preserving spread

prices, the ability to support collusion is independent of the 'distance' of the collusive outcome from either the Nash equilibrium or the joint profit-maximizing outcome of the stage game. In contrast, for the sharing game considered here with linear payoffs and a uniform distribution of innovations, the fully collusive solution (in which $G = \mu$) is the easiest to support, and minimally collusive solutions (in which G is close to 0) are impossible to support. The peculiarity of this result arises from the assumed indivisible nature of the innovations, as we shall see.

If we hold the mean, μ, of the uniform distribution constant and increase the spread parameter, s, we have a mean-preserving spread of the distribution. Such a spread twists the $X(G)/G$ schedule so as to make it easier to support small reciprocal gifts and harder to support large ones. In fact, the derivative of $X(G)/G$ with respect to s is negative or positive as G is smaller or larger, respectively, than $(\mu - s)/2$. (As we argue below, this 'twist' is a general result for the linear payoff model.) Since, for the case of linear payoffs and a uniform distribution, for any sharing agreement to be feasible full sharing must be supportable, it follows that a sufficiently drastic increase in the spread parameter could render unviable all sharing agreements.

In figure 2, we illustrate a mean-preserving spread of the distribution $f(x)$, that produces a new distribution $f'(x)$. The spread transfers mass from $f(x)$ within the interval shown as (x_1, x_2) to the tails of $f'(x)$, while preserving the mean. Let $X'(G)$ denote the largest element of the form-optimal sharing agreement that delivers the reciprocal gift exchange G under distribution $f'(x)$. Clearly, for reciprocal gifts G such that $X'(G) \leq x_1$, it is the case that $X'(G) < X(G)$ because $f'(x) > f(x)$ for innovations $x \leq x_1$. It follows immediately that these 'small' reciprocal gifts are easier to support under $f'(x)$ than under $f(x)$. Conversely, for reciprocal gifts G such that $X(G) \geq x_2$, it is the case that $X'(G) > X(G)$ and these 'large' gifts are easier to support under $f(x)$ than under $f'(x)$. In fact, for this sort of mean-preserving spread, it is easily shown that there is a reciprocal gift $\tilde{G}$ and a corresponding largest element $X(\tilde{G})$ of the associated form-optimal agreement satisfying $x_1 < X(\tilde{G}) < x_2$, such that $X'(G) < X(G)$ for $G < \tilde{G}$ and $X'(G) > X(G)$ for $G > \tilde{G}$. Consequently, a mean-preserving spread generally makes it easier to

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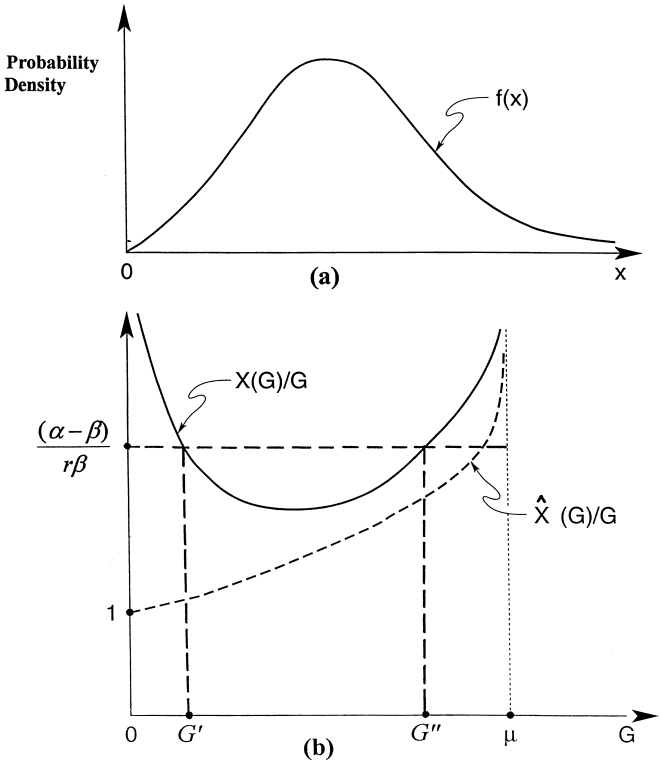


FIGURE 3 The gamma distribution

support relatively small reciprocal gifts and more difficult to support relatively large ones.

We saw in figure 1 (b) that, for the uniform distribution, $X(G)/G$ is downward sloping. However, this is not a general feature of sharing arrangements. To see this, suppose now that the innovations are drawn from a gamma distribution:

$$f(x) = \frac{1}{\Gamma(b)} a^b x^{b-1} e^{-ax} \quad \text{for} \quad x \geq 0, a > 0, b > 0, \tag{25}$$

where $\Gamma(.)$ is the gamma function. The mean and variance of this distribution are b/a and b/a^2, respectively. Figure 3 (a) illustrates the density function associated with the gamma distribution, and the solid line in figure 3 (b) displays the corresponding $X(G)/G$ function. The assumed values for both parameters a and b are equal to 5 in the figure. For these parameter values, the density function rises steeply for very low innovations and falls rapidly for very high innovations, and this feature leads to the U-shape of the $X(G)/G$ function. Consequently, very small and very large reciprocal gifts are both unsupportable; only intermediate valued reciprocal gifts – those in the interval $[G', G'']$ in the figure – are supportable.

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In the example with a uniform distribution, larger reciprocal gifts were easier to sustain than smaller ones, whereas with the gamma distribution we see that this is not necessarily so. Why is there this indeterminacy regarding the ease or difficulty of supporting larger reciprocal gifts? We have seen that the ratio $X(G)/G$ is a measure of the largest benefit-cost ratio of cheating on the given agreement. This ratio needs to be below the critical value given on the right-hand side of (22) if the sharing agreement is to be supportable. If the reciprocal gift is increased by a certain amount, the amount by which $X(G)$ has to increase to generate the required increase in G clearly depends on the value of the density function at $X(G)$, that is, on $f(X(G))$: if this density is large (small), the required increase in $X(G)$ will be small (large). Thus, depending on the value of the density at $X(G)$, the ratio $X(G)/G$ could either increase or decrease as G increases. In other words, larger reciprocal gifts may be either harder or easier to support.

6. Perfectly divisible innovations

To get a deeper understanding of the role of indivisibility in our results, in this section we explore the set of supportable sharing agreements when innovations are *perfectly divisible*. Thus, in any period, a firm can choose to share an arbitrary fraction of its innovation with its rival. The development of the analysis closely parallels that for indivisible innovations.

Given perfect divisibility, a sharing agreement is described by a function $g(x)$ with domain $[\underline{x}, \bar{x}]$ satisfying the restriction that $0 \leq g(x) \leq x$. This is the gift bestowed on its rival by a firm whose innovation is x. A *form-optimal* sharing agreement is characterized by a variable, $\hat{X}$, with $0 < \hat{X} \leq \bar{x}$, and has the following form:

$$g(x) = \begin{cases} x & \text{if } x < \hat{X} \\ \hat{X} & \text{if } x \geq \hat{X}. \end{cases} \tag{26}$$

Under a form-optimal agreement, if a firm's innovation is less than $\hat{X}$, the firm's gift to its competitor is its entire innovation; but if its innovation exceeds $\hat{X}$, the firm's gift to its competitor is equal to $\hat{X}$. When innovations are indivisible, the sharing is inevitably all-or-nothing. Here, in sharp contrast, as long as $\hat{X} > 0$ the agreement always calls for *some* sharing, regardless of the size of the firm's innovation.

When the payoff function is linear in know-how, it can readily be shown that agreements that are not form optimal can be (weakly) Pareto-dominated by a form-optimal agreement. However, this cannot be demonstrated for general functional forms. Nevertheless, in what follows we shall focus only on form-optimal sharing agreements – which we denote by $\hat{\Omega}(G)$ – and we shall characterize them by the maximal element, $\hat{X}(G)$, of the set $\hat{\Omega}(G)$ associated with the reciprocal gift, G. There is a one-to-one relationship between $\hat{X}(G)$ and G, and the former is that value of X that solves the following:

$$G = \int_{\underline{x}}^{\hat{X}} xf(x)\,dx + \int_{\hat{X}}^{\bar{x}} \hat{X} f(x)\,dx. \tag{27}$$

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Similarly, for any X the implied reciprocal gift exchange, $\hat{G}(X)$, is given by the right-hand side of (27). Note that in the case of indivisible innovations, the maximal element, $X(G)$, of the agreement set associated with any positive G, however small, was bounded below by $\underline{x}$. In contrast, in the case of perfectly divisible innovations, $\hat{X}(G)$ can take on any value in the interval $[0, \bar{x}]$, depending on the value of G.

The set $\hat{\Omega}(G)$ is defined by

$$\hat{\Omega}(G) \equiv \{x : \underline{x} \leq x \leq \hat{X}(G)\},$$

and the complement of $\hat{\Omega}(G)$ will be denoted by the set $\hat{\Lambda}(G)$. By construction, an agreement that requires a firm to share all of its innovations if it falls within $\hat{\Omega}(G)$, and an amount $\hat{X}(G)$ if it falls in $\hat{\Lambda}(G)$ will generate a reciprocal gift exchange equal to G. If $G < \underline{x}$, then by (27) we must have $\hat{X}(G) = G$. In this case, $\hat{\Omega}(G)$ is the null set Φ and $\hat{\Lambda}(G)$ is Υ.

The expressions for the expected ex ante profit per period, $E\hat{\Pi}^i_{\hat{\Omega}}$, and the temptation to cheat, $\hat{T}^i_{\hat{\Omega}}(x_i)$, in the case of perfect divisibility of innovations are, of course, different from those under indivisibility but are readily written down. Suppressing the arguments of $\hat{X}(G), \hat{\Omega}(G)$, and $\hat{\Lambda}(G)$ for brevity, if $\underline{x} \leq \hat{X} \leq \bar{x}$, for example,

$$E\hat{\Pi}^i_{\hat{\Omega}} = \langle \Pi^i(x_i, x_j; x_j, x_i)\rangle_{\hat{\Omega},\hat{\Omega}} + \langle \Pi^i(x_i, x_j; \hat{X}, x_i)\rangle_{\hat{\Omega},\hat{\Lambda}} + \langle \Pi^i(x_i, x_j; x_j, \hat{X})\rangle_{\hat{\Lambda},\hat{\Omega}} + \langle \Pi^i(x_i, x_j; \hat{X}, \hat{X})\rangle_{\hat{\Lambda},\hat{\Lambda}}, \tag{28}$$

and if, in addition, $x_i \leq \hat{X}$,

$$\hat{T}^i_{\hat{\Omega}}(x_i) = \int_{\underline{x}}^{\hat{X}} \Psi^i(x_i, x_j; x_j) f(x_j)\, dx_j + \int_{\hat{X}}^{\bar{x}} \Psi^i(x_i, x_j; \hat{X}) f(x_j)\, dx_j. \tag{29}$$

The temptation to cheat when $x_i > \hat{X}$ can be analogously written down. If, on the other hand, $0 \leq \hat{X} \leq \underline{x}$, then

$$E\hat{\Pi}^i_{\hat{\Omega}} = \langle \Pi^i(x_i, x_j; \hat{X}, \hat{X})\rangle_{\Upsilon,\Upsilon}, \tag{30}$$

and

$$\hat{T}^i_{\hat{\Omega}}(x_i) = \int_{\underline{x}}^{\bar{x}} [\Pi^i(x_i, x_j; \hat{X}, 0) - \Pi^i(x_i, x_j; \hat{X}, \hat{X})] f(x_j)\, dx_j. \tag{31}$$

For the agreement set $\hat{\Omega}(G)$ to be supportable as a SPNE, the following inequality must hold for all $x_i \in \Upsilon$:

$$\hat{P}^i_{\hat{\Omega}}(\delta) \geq \hat{T}^i_{\hat{\Omega}}(x_i), \tag{32}$$

where the left-hand side is the expected punishment for cheating, given by

$$\hat{P}^i_{\hat{\Omega}}(\delta) = \frac{\delta}{1-\delta}[E\hat{\Pi}^i_{\hat{\Omega}} - E\Pi^i_{\Phi}], \tag{33}$$

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with $E\Pi^i_\Phi$ denoting, as before, the expected ex ante payoff per period to firm i when no sharing agreement is in place.

Note that for full sharing (that is, $G = \mu$ or $\hat{\Omega}(G) = \Upsilon$), we must have $\hat{X} = \bar{x}$; so the necessary and sufficient condition for full sharing is the same for perfect divisibility as for the indivisibility case. (When there is to be full sharing, it does not matter whether innovations are divisible or indivisible, since the agreements call for the sharing of every innovation in full.) The following proposition looks at sharing possibilities for small reciprocal gift exchanges when innovations are perfectly divisible.

PROPOSITION 3. *When innovations are perfectly divisible, there exists a sufficiently large discount factor bounded away from one above which sharing agreements with small gift exchanges are always supportable.*

To prove this result, note that when $G \to 0$, from (27) it follows that $\hat{X}(G) \to 0$. Thus, if $\underline{x} > 0$, the relevant scenario is that with $0 < \hat{X} \leq \underline{x}$. Expanding the integrand in (30) in a Taylor series around $\hat{X}(G) = 0$ and retaining only terms upto the first order in $\hat{X}(G)$, we obtain

$$E\hat{\Pi}^i_{\hat{\Omega}} = E\Pi^i_\Phi + \hat{X}(G)\gamma, \tag{34}$$

where

$$\gamma \equiv \int_{\underline{x}}^{\bar{x}} \int_{\underline{x}}^{\bar{x}} [\Pi^i_3(x_i, x_j; 0, 0) + \Pi^i_4(x_i, x_j; 0, 0)] h(x_i, x_j)\, dx_i\, dx_j, \tag{35}$$

in which the subscripts of the payoff functions, as before, denote derivatives with respect to the respective arguments. Note that, using inequality (2), it immediately follows that $\gamma > 0$. From (33) and (34), it follows that the punishment for cheating on small agreements is given by

$$\hat{P}^i_{\hat{\Omega}}(\delta) = \frac{\delta}{1-\delta}\hat{X}(G)\gamma. \tag{36}$$

Similarly, expanding the integrand in (31) in a Taylor series around $\hat{X}(G) = 0$ and retaining only terms up to the first order in $\hat{X}(G)$, we obtain

$$\hat{T}^i_{\hat{\Omega}}(x_i) = \hat{X}(G) S^i(x_i), \tag{37}$$

where

$$S^i(x_i) \equiv \int_{\underline{x}}^{\bar{x}} [-\Pi^i_4(x_i, x_j; 0, 0)] f(x_j)\, dx_j \tag{38}$$

$$> 0,$$

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using the last inequality in (1). Using (36), (37), and (38) in (32), the necessary and sufficient incentive compatibility condition that must be satisfied for agreements with very small gift exchanges to be supportable as SPNE can be written

$$\frac{\delta}{1-\delta}\,\gamma \geq \max_{x_i} S^i(x_i). \tag{39}$$

Since the right-hand side is finite, there is always a δ sufficiently close to (but strictly smaller than) unity for which the above inequality is satisfied. This establishes the proposition.

The difference between proposition 2 for indivisible innovations and proposition 3 for perfectly divisible ones is that the discount factor required to support agreements with arbitrarily small reciprocal gifts, G, is unity in the limit when $G = 0$. In the case of perfectly divisible innovations, however, the required discount factor remains strictly bounded away from unity even in the limit when $G = 0$.

To get an intuitive grasp of proposition 3, we need the converse of the aphorism we used to discuss proposition 2: 'A bird in the hand is never worth many in the bush, when the birds in the bush are caught with high probability.' Again, the bird in the hand is $\hat{X}(G)$, the largest gift the firm must share with its competitor if the sharing agreement is to be supportable, and the birds in the bush are the gifts that the competitor will share in the future with the firm if the sharing agreement is in place. When G is small, in each future period, there is a high probability that the competitor will bestow a gift equal to $\hat{X}(G)$ because most innovations will exceed $\hat{X}(G)$, and this probability approaches 1 as G approaches zero. Hence, for all practical purposes, not sharing $\hat{X}(G)$ today means foregoing a gift of the same magnitude in most future periods. According to proposition 3, as long as the discount factor is large enough to ensure that (39) is satisfied, the birds in the bush are worth more than the bird in the hand when G is small.

6.1. The example with linear payoffs again

Pursuing the earlier example with linear payoffs, we note that when innovations are perfectly divisible a form-optimal agreement $\hat{\Omega}(G)$ is supportable as an SPNE if and only if

$$\frac{\hat{X}(G)}{G} \leq \frac{\delta}{1-\delta}\,\frac{(\alpha-\beta)}{\beta}. \tag{40}$$

When the innovations are distributed uniformly according to (24), it is readily verified that

$$\hat{X}(G)/G = \begin{cases} [(\mu+s) - 2s^{1/2}(\mu-G)^{1/2}]/G & \text{if } G \geq \mu - s, \\ 1 & \text{if } G < \mu - s. \end{cases} \tag{41}$$

This function is illustrated by the dashed curve in figure 1 (b). When G is equal to μ, the values of $\hat{X}(G)/G$ and $X(G)/G$ are identical, and when G is less than μ, $\hat{X}(G)/G$ lies below $X(G)/G$. As noted earlier, under the assumption of perfect

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divisibility, even though the smallest innovation is $\underline{x}$, the amount of information to be shared can be smaller than $\underline{x}$. We have noted earlier that for $G \leq \underline{x}$, we necessarily have $\hat{X}(G)/G = 1$. With the uniform distribution and divisible innovations, the necessary and sufficient condition for small reciprocal gifts to be supportable is $1 \leq \delta(\alpha - \beta)/[\beta(1 - \delta)]$. This is a special case of the general condition (39), since in this linear payoff case $\gamma = (\alpha - \beta)$ and $S^i(x_i) = \beta$ for all x_i.

A feature worth noting is that it is never easier to support larger gifts than smaller ones because, as can be seen from figure 1 (b), the slope of $\hat{X}(G)/G$ is non-negative. This feature is now shown to obtain for arbitrary density functions. Differentiating the function $\hat{X}(G)/G$ with respect to G reveals that this function is increasing in G for $G > \underline{x}$ if $d\hat{X}(G)/dG > \hat{X}(G)/G$ or, equivalently, if $\hat{G}(X)/X > d\hat{G}(X)/dX$. But from (27),

$$\frac{d\hat{G}(X)}{dX} = 1 - F(X),$$

where $F(.)$ is the cumulative distribution function associated with $f(.)$, and

$$\hat{G}(X)/X = (1/X)\int_{\underline{x}}^{X} xf(x)\,dx + 1 - F(X)$$

$$> \frac{d\hat{G}(X)}{dX}.$$

Hence, for $G > \underline{x}$, $\hat{X}(G)/G$ is an increasing function of G. Furthermore, we have seen above that for $G \leq \underline{x}$, $\hat{X}(G)/G = 1$. This establishes our last result, stated below.

PROPOSITION 4. *When innovations are perfectly divisible and the payoff function is linear, it is never easier to support larger reciprocal gifts than smaller.*

In figures 1 (b) and 3 (b), respectively, the dashed curves display the function $\hat{X}(G)/G$ for the uniform and gamma distributions introduced earlier. The non-negative slope of this function confirms, for these special cases, the validity of the result in proposition 4.

When innovations were indivisible we have seen that, depending on the probability distribution for innovations, larger reciprocal gift exchanges may be easier or harder to support than smaller ones. Why is it that, with divisible innovations, larger reciprocal gift exchanges are never easier to support than smaller ones? When innovations are divisible, a firm is called upon to share $\hat{X}(G)$ even when its realization exceeds $\hat{X}(G)$, and this occurs with a probability $1 - F(\hat{X}(G))$, where $F(.)$ is the cumulative distribution function. When the reciprocal gift exchange is increased by a small amount ΔG, the increase $\Delta\hat{X}$ in $\hat{X}(G)$ needed to generate this increase in G is simply given by $[1 - F(\hat{X}(G))]\Delta\hat{X} = \Delta G$. Since the probability in the square bracket is never greater than unity, it follows that $\Delta\hat{X}$ must be at least as large as ΔG. This means that the ratio $\hat{X}(G)/G$, which measures the maximum temptation to cheat relative to the punishment, cannot decline when G increases.

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According to this proposition, firms in equilibrium are likely to share small innovations but not very large ones. This is because large innovations come infrequently and the sharing of such an innovation is not likely to be reciprocated soon. The above proposition is consistent with a liberal interpretation of some anecdotal evidence from the aerospace industry. von Hippel (1988, 86) reports that, according to an aerospace engineer, 'they freely exchange most know-how under normal circumstances. But, when competition for an important government contract is in the offing, the situation changes, and trading of information between rivals that might affect who wins the contract stops. Later, after the contract has been awarded, the same know-how that was recently closely guarded will apparently again be traded.' In other words, know-how that is potentially of high value is witheld.[6]

7. Conclusions

In this paper, in a repeated game framework, we have modelled the phenomenon of know-how sharing between two firms when they stochastically realize innovations of various sizes. We separately considered the cases in which the innovations are divisible and those in which they are indivisible, a distinction that turns out to be important. We have shown that both stochasticity and indivisibility of innovations inhibit sharing. We have demonstrated that, in equilibrium, sharing may be observed to be intermittent even when the sharing agreement has not been violated. When innovations are divisible, we have found that small innovations are more likely to be shared than larger ones. When innovations are indivisible, however, we have shown that agreements to share only small innovations are never supportable.

Inspired by Axelrod's (1984) book, a great deal has been written in the past fifteen years about the possibilities of achieving cooperation in the iterated Prisoners' Dilemma (IPD) game. The problem is significant because many – perhaps most – social and economic interactions have the flavour of the IPD game. If the players are not careful to devise clever strategies in these games, they will find themselves locked in a low-level Nash equilibrium trap. Our results suggest that, in the realm of know-how sharing, one cannot be overly optimistic about the prospects for cooperation. If the stochastic innovations of firms are indivisible – in which case, sharing is an all-or-nothing proposition – even the sharing of small innovations cannot easily be supported as subgame perfect Nash equilibria. There is room for optimism only for the case in which the innovations are divisible, a feature that is not generic to many scenarios. Ideas, for example, are often indivisible; one cannot easily share half an idea. Thus, in the realm of know-how sharing between competing firms, for example, the indivisibility of ideas or innovations militates against the possibility of cooperation. Given the fact that knowledge is a non-rivalrous public good, these results suggest that governments might consider eliminating the menace of anti-trust action against competing firms that trade in know-

6 Though this quote seems to contradict one of our assumptions of convenience (namely, that firms cannot pass on know-how that they had earlier witheld), its thrust is nevertheless consistent with the spirit of our analysis.

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how. Otherwise, since tacit agreements are unenforceable in the courts, the public interest may not be well served.

In this paper, we took the firms to be identical. This simplified the analysis, since we could conduct our investigation by examining only one incentive compatibility constraint. We briefly remark here on the effects of asymmetry on sharing possibilities. Suppose we render the firms asymmetric by reducing the discount factor of one of them. Their incentive compatibility constraints would differ now, but the binding one would be that of the firm with the lower discount factor (since it is this firm that would be more tempted to cheat). Clearly, the greater impatience of this firm would undermine the extent of sharing. What is more interesting, we could also speculate on the effect of introducing asymmetry in the distribution of the innovations. Suppose we reduce the expected value of innovations for one firm but hold the variance constant. We would expect the distribution of this firm to be the constraining factor in limiting the extent of sharing, since its rival would be less willing than before to part with its information. For small asymmetries sharing would still be possible, especially if information is divisible. In the presence of extreme asymmetry, however, sharing is likely to break down. In general, since sharing presumes reciprocity, asymmetry in the distribution of innovations would tend to militate against the possibility of sharing.

The more differentiated the products that firms produce, the less know-how trading we would expect. This is because the know-how of such firms is more specialized. Know-how that these firms would willingly trade is probably of little use to others, whereas know-how that is particularly important to a firm is likely to be of the kind that differentiates this firm from others and so will likely be witheld. This would explain why one of the firms (Quanex) in von Hippel's (1988) study of the steel minimill industry did not engage in know-how trading. Unlike the other firms in his sample, Quanex produced specialty steel products.

We now briefly discuss an issue we did not raise in the paper so far, namely, that of renegotiation. There are really two issues here. The first is whether the optimal sharing agreement, which we have taken to be independent of time, is really so, or whether the firms will want to renegotiate the agreement sets as the game proceeds. The answer to this is immediate: the sharing agreement deemed optimal by the firms will not change from period to period, since the game considered here is stationary.[7] Secondly, as is frequently the case in repeated games, the game considered here is open to a more standard objection. In the event of cheating, the firms might consider renegotiating in order to avoid the costly no-sharing punishment phase, and this possibility would undermine their attempts at sharing information. We do not address this issue here. We may point out, however, that our earlier work (Eaton and Eswaran 1997), where we considered a scenario in which n firms ($n \geq 3$) trade technologies in a supergame, is relevant to this issue. We argued that cheating on a technology-sharing agreement may be interpreted by the non-

7 Even in the model with stocks, alluded to in footnote 1, this remains true. It will also be true in the event the firms are asymmetric. If the degree of asymmetry changes over time, however, the optimal sharing agreement will also change.

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cheaters as a display of bad faith by the cheater and that they might subsequently eject the cheater from the trading coalition while continuing to trade technologies among themselves. We demonstrated that pursuing this strategy instead of dissolving the coalition in the event of cheating by a member not only would be in the self-interest of the honest members but also would inflict the largest punishment on the cheater. While this does not render the supergame equilibrium renegotiation proof, it is less open to this objection than is the standard presumption that cooperation is completely discontinued in the event of cheating. In future research, a similar approach might be worthy of consideration in the context of the present paper. However, the thrust of our results is likely to be robust: indivisibility and stochasticity render information less conducive to sharing in repeated games.

Finally, we observe that the degree of cooperation between firms, in general, might be expected to vary from time to time. Our model suggests two possible reasons for this. In a form optimal agreement, firms see in advance that the sharing of really important and rare ideas and innovations cannot be supported. Hence, in those periods when such innovations occur, the lucky firm will not share its good fortune with its competitor. To see the other explanation, suppose, for simplicity, that firms have inelastic expectations with respect to the interest rate. The set of supportable sharing agreements will then be a function of the interest rate and, in any circumstance where full sharing is not supportable but partial sharing is supportable, a decrease in the interest rate will unambiguously increase the set of supportable sharing agreements. This is because the punishments for defection that occur in future periods are given a higher weight in the firms' intertemporal profit calculation. Then, presuming that firms pick the most profitable sharing agreement for each interest rate, we should expect to see more sharing on average in those periods in which the interest rate is low.

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